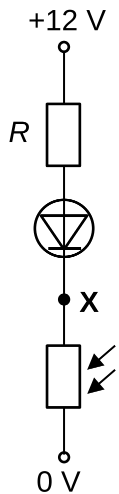
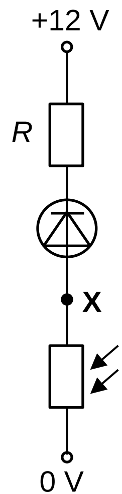
22 In bright light, a light-dependent resistor (LDR) has a resistance of R. It is connected in series with an ideal diode and a fixed resistor of resistance R. An ideal diode has zero resistance in the forward direction and infinite resistance in the reverse direction.

In which arrangement will the potential at **X** increase when the circuit is moved to a darker environment?

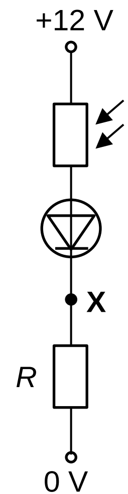
A



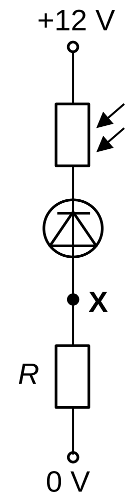
B



C



D



L2	**Answer: A** Diodes in **options B and D** are in **reverse biased** connection (**like an open circuit where the diode is**). → **No current flows → zero p.d. across the resistance → potential at X = 0 V in both bright and dark conditions**, i.e. **no change in potential at X** for options B and D. → Eliminate options B and D. Diodes in **options A and C** are in **forward biased** connection (**like zero resistance where the diode is**). → **Current flows → non-zero p.d. across the resistance**. Since **LDR's resistance increases when moved into the dark**, by Potential Divider Principle, the **p.d. across the LDR will increase**. **Hence in option A potential at X will increase, while in option C potential at X will decrease.**

23	A rotary potentiometer consists of a length of uniform resistance wire connected to the terminals